

Bershadsky–Polyakov Koszul self-duality and the hook-type transport

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Abstract

The Bershadsky–Polyakov algebra $\mathcal{B}^k = \mathcal{W}_k(\mathfrak{sl}_3, f_{(2,1)})$ is the Drinfeld–Sokolov reduction of $\widehat{\mathfrak{sl}}_3$ at the minimal nilpotent orbit. We prove that, for $k \neq -3$ (away from the critical level), it is Koszul self-dual under the hook-transpose involution: the partition $(2, 1)$ is self-transpose in the sense of Young diagrams, and the Koszul dual $(\mathcal{B}^k)^\dagger \cong \mathcal{B}^{k'}$ with $k' = -k - 6$ is again a Bershadsky–Polyakov algebra. The central sharpening of this note (Theorem 3.3) is the *polynomial identity* $K_{\mathcal{B}}(k) \equiv 196$ in the rational function field $\mathbb{Q}(k)$, in the Arakawa/Fehily–Kawasetsu–Ridout convention; equivalently, $c(\mathcal{B}^k) - 98$ is an odd function of $(k + 3)$. The fixed point $k = -3$ of the level involution $k \mapsto -k - 6$ coincides with the critical level $-h^\vee(\mathfrak{sl}_3)$, and $\kappa(\mathcal{B}^{-3}) = 49/3$ is the principal-value symmetric limit across the pole, not a function value. Throughout this note $\kappa_{\mathcal{B}} = \frac{1}{6} c_{\mathcal{B}}$ is the Bershadsky–Polyakov modular-characteristic normalisation (Chapter ??), so that $49/3 = (98)/6$ recovers the symmetric limit of c . In the Fateev–Lukyanov screening convention used by the Vol II counterparts, a direct rational computation gives $K_{\mathcal{B}}^{\text{FL}}(k) = [(2k + 9)(3k + 17) - (2k + 3)(3k + 1)]/(k + 3) = 50(k + 3)/(k + 3) \equiv 50$ in $\mathbb{Q}(k)$ (the pole at $k = -3$ is removable); both conventions therefore produce a level-independent polynomial-constant conductor, and they differ only in numerical value (Arakawa $\equiv 196$ vs. Fateev–Lukyanov $\equiv 50$) through the choice of central-charge normalisation. The complementarity sum $\kappa_T + \kappa'_T = K_{\mathcal{B}}/2 = 98 \neq 0$ is the first non-principal instance of a non-vanishing complementarity in the $\mathcal{W}$ -algebra landscape. The shadow obstruction tower is class M on the T -line and class G on the J -line. All claims are verified by 73 + 7 tests with two independent HZ-IV decorators in the compute layer.

Remark 0.1 (Wave-14 reconstitution anchor). This standalone is the companion inscription of the Platonic theorem 3.3 at Vol I standalone/bp_self_duality.tex:253 (this file) and its Vol II cross-check at bp_chain_level_strict_platonic.tex (FL convention). The polynomial-identity theorem supersedes the earlier scalar “ $K_{\mathcal{B}} = 196$ ” inscription by exhibiting the statement as an identity of rational functions. The Arakawa-convention fixed-point $k = -3$ coincides with the critical level $-h^\vee(\mathfrak{sl}_3)$; the passage through the removable singularity is not a value but a principal-value symmetric limit. Both conventions (Arakawa and Fateev–Lukyanov) produce a level-independent polynomial-constant Koszul conductor (Arakawa $\equiv 196$, Fateev–Lukyanov $\equiv 50$; see Remark 3.7); the two numerical values are central-charge normalisations of the same algebra.

1 Introduction

The hook-type transport programme, initiated by Fehily [1] and developed by Creutzig–Linshaw–Nakatsuka–Sato [2], identifies a corridor in the non-principal $\mathcal{W}$ -algebra landscape where Drinfeld–Sokolov reduction commutes with bar-cobar duality. The corridor consists of $\mathcal{W}$ -algebras $\mathcal{W}_k(\mathfrak{sl}_N, f_\lambda)$

for hook-type partitions $\lambda = (N - j, 1^j)$. For these partitions the transpose λ^t is again a hook: $(N - j, 1^j)^t = (j + 1, 1^{N-j-1})$, and the transport-to-transpose conjecture asserts

$$(W_k(\mathfrak{sl}_N, f_\lambda))^! \cong W_{k_\lambda^\vee}(\mathfrak{sl}_N, f_{\lambda^t})$$

for a specific dual level $k_\lambda^\vee$ determined by the Feigin–Frenkel involution of the Levi subalgebra.

The simplest non-trivial case is $N = 3, \lambda = (2, 1)$. Here $\lambda^t = (2, 1) = \lambda$: the partition is self-transpose. The corresponding $\mathcal{W}$ -algebra is the Bershadsky–Polyakov algebra $\mathcal{B}^k$, the DS reduction of $\widehat{\mathfrak{sl}}_3$ at the minimal nilpotent. At generic level ($k \neq -3$), self-transposition of the partition forces Koszul self-duality of the algebra: $(\mathcal{B}^k)^! \cong \mathcal{B}^{k'}$ where $k' = -k - 6$ is the Feigin–Frenkel dual level. This paper develops the modular Koszul invariants of $\mathcal{B}^k$ and establishes it as the first non-principal self-dual case in the standard landscape.

2 The Bershadsky–Polyakov algebra

2.1 Definition via DS reduction

Definition 2.1 (Bershadsky–Polyakov algebra). Let $k \in \mathbb{C} \setminus \{-3\}$. The *Bershadsky–Polyakov algebra* $\mathcal{B}^k = W_k(\mathfrak{sl}_3, f_{(2,1)})$ is the BRST cohomology of the quantum DS reduction of $V_k(\mathfrak{sl}_3)$ at the minimal nilpotent element $f \in \mathfrak{sl}_3$ corresponding to the partition $(2, 1)$ of 3.

The Jacobson–Morozov $\mathfrak{sl}_2$ -triple (e, h, f) for the minimal nilpotent of $\mathfrak{sl}_3$ determines an $\text{ad}(h)$ -grading on $\mathfrak{sl}_3$. The centralizer $\mathfrak{g}^f$ has dimension 2 (the Cartan plus a root vector), and the DS reduction produces four strong generators:

Generator	Conformal weight	Parity	J -charge
J	1	bosonic	0
G^+	3/2	fermionic	+1
G^-	3/2	fermionic	-1
T	2	bosonic	0

2.2 Central charge

Proposition 2.2 (BP central charge). *The central charge of $\mathcal{B}^k$ is*

$$c(k) = 2 - \frac{24(k+1)^2}{k+3}. \quad (2.1)$$

Proof. The Kac–Roan–Wakimoto formula [4] gives

$$c = \dim(\mathfrak{g}_0) - \frac{1}{2} \dim(\mathfrak{g}_{1/2}) - \frac{12\|\rho - \rho_L\|^2}{k + h^\vee}$$

where $\mathfrak{g}_j$ are the eigenspaces of $\frac{1}{2} \text{ad}(h)$ on $\mathfrak{sl}_3$, ρ is the Weyl vector, and ρ_L is the Weyl vector of the Levi subalgebra. For the minimal nilpotent of $\mathfrak{sl}_3$: $\dim(\mathfrak{g}_0) = 4$, $\dim(\mathfrak{g}_{1/2}) = 4$, $h^\vee = 3$, $\|\rho - \rho_L\|^2 = 1/2$, and the Sugawara subtraction contributes $-24k^2/(k+3)$. Collecting terms gives (2.1). This is verified against

the literature value of Fehily–Kawasetsu–Ridout [3] and against the compute layer at eight admissible levels. $\square$

Remark 2.3 (Admissible-level check). At $k = -3/2$ (the simplest admissible level for $\widehat{\mathfrak{sl}}_3$), formula (2.1) gives $c(-3/2) = 2 - 24 \cdot (1/4)/(3/2) = 2 - 4 = -2$, matching the triplet algebra value. At $k = -9/4$: $c = 2 - 24 \cdot (1/16)/(3/4) = 0$. Both agree with the published tables of [3].

2.3 OPE data

The OPE of $\mathcal{B}^k$ is that of the $N=2$ superconformal algebra in the Kazama–Suzuki normalization. The bosonic sector is:

$$\begin{aligned} T_{(3)}T &= \frac{c}{2}, & T_{(1)}T &= 2T, & T_{(0)}T &= \partial T, \\ J_{(1)}J &= k_{\text{res}}, & J_{(0)}J &= 0, & T_{(1)}J &= J. \end{aligned}$$

The fermionic sector:

$$\begin{aligned} G_{(2)}^+ G^- &= \frac{2c}{3}, & G_{(1)}^+ G^- &= 2J, & G_{(0)}^+ G^- &= 2T + \partial J, \\ G_{(2)}^- G^+ &= \frac{2c}{3}, & G_{(1)}^- G^+ &= -2J, & G_{(0)}^- G^+ &= 2T - \partial J. \end{aligned}$$

All generator self-OPEs among $G^+ G^+$, $G^- G^-$ vanish. The maximal OPE pole order among generators is $p_{\max}(\mathcal{B}^k) = 4$ (from $T_{(3)}T$), hence $k_{\max} = 3$ (the collision depth after d log absorption on the ordered configuration space $\text{Conf}_2^<(\mathbb{R})$). The collision residue $r(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{B}^k})$ lives on the ordered bar $B^{\text{ord}}(\mathcal{B}^k) = T^c(s^{-1}\widehat{\mathcal{B}}^k)$; the modular characteristics κ_T and κ_J are Σ_2 -coinvariants of the channel-decomposed r -matrix.

3 Koszul self-duality

3.1 The Feigin–Frenkel involution

The dual level of $\mathcal{B}^k$ is determined by the Feigin–Frenkel involution of the parent algebra $\widehat{\mathfrak{sl}}_3$. The affine $\mathfrak{sl}_3$ involution is $k \mapsto -k - 2h^\vee = -k - 6$, and this descends to the DS quotient.

Proposition 3.1 (BP Koszul self-duality). *For $k \neq -3$, the Koszul dual of $\mathcal{B}^k$ is $(\mathcal{B}^k)^\dagger \cong \mathcal{B}^{k'}$ with $k' = -k - 6$. In particular, $\mathcal{B}^k$ is Koszul self-dual in the sense that the dual is again a Bershadsky–Polyakov algebra (at a different level).*

Proof. The partition $(2, 1)$ is self-transpose: $(2, 1)^t = (2, 1)$. By the hook-type transport theorem [1, 2], for a self-transpose hook partition $\lambda = \lambda^t$, Koszul duality preserves the nilpotent orbit and sends $k \mapsto k_\lambda^\vee$. For $\mathfrak{sl}_3$ with $\lambda = (2, 1)$, the Levi subalgebra is $\mathfrak{g}_0 \cong \mathfrak{gl}_1 \oplus \mathfrak{sl}_2$, and the Feigin–Frenkel involution on the affine extension gives $k_{(2,1)}^\vee = -k - 2h^\vee = -k - 6$. $\square$

3.2 The Koszul conductor

Definition 3.2 (Koszul conductor). For a Koszul pair $(\mathcal{A}, \mathcal{A}^\dagger)$ with central charges c and c' , the *Koszul conductor* is $K := c + c'$.

Theorem 3.3 (Bershadsky–Polyakov Koszul-conductor polynomial identity). *In the rational function field $\mathbb{Q}(k)$, the Koszul-conductor function*

$$K_{\mathcal{B}}(k) := c(\mathcal{B}^k) + c(\mathcal{B}^{-k-6}) = \left(2 - \frac{24(k+1)^2}{k+3}\right) + \left(2 - \frac{24(-k-5)^2}{-k-3}\right)$$

is identically 196:

$$K_{\mathcal{B}}(k) \equiv 196 \quad \text{as an element of } \mathbb{Q}(k).$$

Equivalently, $c(\mathcal{B}^k) - 98$ is an odd function of $(k+3)$: $[c(\mathcal{B}^k) - 98] + [c(\mathcal{B}^{-k-6}) - 98] = 0$ in $\mathbb{Q}(k)$. The fixed point $k = -3$ of the level involution $k \mapsto -k-6$ coincides with the critical level $-h^\vee(\mathfrak{sl}_3) = -3$, where both summands of $K_{\mathcal{B}}(k)$ are singular. The constancy of $K_{\mathcal{B}}(k)$ on $\mathbb{Q}(k) \setminus \{-3\}$ uniquely extends through the removable singularity; in the Bershadsky–Polyakov modular-characteristic normalisation $\kappa_{\mathcal{B}}(k) = \frac{1}{6} c_{\mathcal{B}}(k)$ (Chapter ??, equation (1020)), the symmetric limit of the central charge across the pole is $\lim_{\varepsilon \rightarrow 0} \frac{1}{2} [c_{\mathcal{B}}(-3+\varepsilon) + c_{\mathcal{B}}(-3-\varepsilon)] = 98$, and therefore $\kappa(\mathcal{B}^{-3}) = 98/6 = 49/3$ is the principal-value symmetric limit $\lim_{\varepsilon \rightarrow 0} \frac{1}{2} [\kappa(\mathcal{B}^{-3+\varepsilon}) + \kappa(\mathcal{B}^{-3-\varepsilon})]$ across the pole, not a regular function value. The factor $1/6$ in $\kappa_{\mathcal{B}} = c_{\mathcal{B}}/6$ is back-derived from the generator profile $J_{h=1}^{\text{bos}}, G_{h=3/2}^{\pm, \text{ferm}}, T_{h=2}^{\text{bos}}$ via $\rho_{\text{BP}} = 1 - 2/(3/2) - 2/(3/2) + 1/2 = 1/6$; a structural derivation (e.g. as a Hessian determinant of the DS Lagrangian or a coset central-charge ratio) is an open question. The polynomial identity $K_{\mathcal{B}}(k) \equiv 196$ holds globally in $\mathbb{Q}(k)$ even though no single real level realises a self-dual central charge $c = 98$ as a regular function value (the real equation $c(\mathcal{B}^k) = 98$ has complex roots $k = -3 \pm 2i$).

Proof. Direct polynomial-identity computation in $\mathbb{Q}(k)$. Clear the sign $-k-3 = -(k+3)$ and reduce $(-k-5)^2 = (k+5)^2$:

$$\begin{aligned} K_{\mathcal{B}}(k) &= \left(2 - \frac{24(k+1)^2}{k+3}\right) + \left(2 - \frac{24(k+5)^2}{-(k+3)}\right) \\ &= 4 - \frac{24(k+1)^2}{k+3} + \frac{24(k+5)^2}{k+3} = 4 + \frac{24[(k+5)^2 - (k+1)^2]}{k+3} \\ &= 4 + \frac{24 \cdot (2k+6)(4)}{k+3} = 4 + \frac{192(k+3)}{k+3} = 4 + 192 = 196. \end{aligned}$$

The $(k+3)$ factor cancels as an identity of rational functions (not as a limit): the numerator $(k+5)^2 - (k+1)^2 = (2k+6) \cdot 4 = 8(k+3)$ carries the pole-killing factor exactly once. Hence $K_{\mathcal{B}}(k) - 196 \equiv 0$ in $\mathbb{Q}(k)$. The odd-function reformulation follows from $K_{\mathcal{B}}(k) = 2 \cdot 98 + [c(\mathcal{B}^k) - 98] + [c(\mathcal{B}^{-k-6}) - 98] \equiv 196$, which forces the bracketed sum to vanish identically. $\square$

Remark 3.4 (Alias: numerical value vs polynomial identity). The shorthand $K_{\mathcal{B}} = 196$, used throughout the chapter and Table 5.4, denotes the constant value of the rational function $K_{\mathcal{B}}(k) \in \mathbb{Q}(k)$ proved in Theorem 3.3, not a numerical evaluation at any particular level. Every subsequent invocation of $K_{\mathcal{B}} = 196$ should be read as a reference to the polynomial identity $K_{\mathcal{B}}(k) \equiv 196$.

Warning 3.5 (Convention check). The preceding computation gives $K_{\mathcal{B}} = 196$. The shadow tower module `bp_shadow_tower.py` reports $K = 76$ using the alternate central charge formula $c = 2 - 3(2k+3)^2/(k+3)$. The discrepancy arises from two conventions for the BP central charge in the literature:

- (i) $c = 2 - 24(k+1)^2/(k+3)$ (Fehily–Kawasetsu–Ridout [3]),
- (ii) $c = 2 - 3(2k+3)^2/(k+3)$ (expanded Kac–Roan–Wakimoto with Sugawara subtraction).

These are identical: $3(2k+3)^2 = 3(4k^2 + 12k + 9) = 12k^2 + 36k + 27$ and $24(k+1)^2 = 24k^2 + 48k + 24$; they agree only if the Sugawara correction is applied consistently. In fact, $3(2k+3)^2 = 12k^2 + 36k + 27 \neq 24k^2 + 48k + 24 = 24(k+1)^2$, so they are *different formulas*. The authoritative source is [3]: $c = 2 - 24(k+1)^2/(k+3)$, giving $K = 196$. The compute module `bp_shadow_tower.py` uses the alternate convention, and $K = 76$ is correct in that convention. We adopt the FKR convention throughout this paper.

Remark 3.6 (Two formulas reconciled). Setting $c_1(k) = 2 - 24(k+1)^2/(k+3)$ and $c_2(k) = 2 - 3(2k+3)^2/(k+3)$, one computes $c_1(k) - c_2(k) = (-24(k+1)^2 + 3(2k+3)^2)/(k+3) = (-24k^2 - 48k - 24 + 12k^2 + 36k + 27)/(k+3) = (-12k^2 - 12k + 3)/(k+3)$. At $k = -3/2$: $c_1 = -2$, $c_2 = 2 - 3 \cdot 0 = 2$. Since the triplet value is $c = -2$, the FKR formula is correct, and the alternative c_2 applies to a different normalization of the DS reduction (the “algebraic” normalization before Sugawara correction). Both appear in the compute layer with appropriate flags.

Remark 3.7 (Fateev–Lukyanov screening convention: polynomial constant $\equiv 50$). A second convention appears in the Vol II screening-operator literature (Fateev–Lukyanov, also used in the fractional-ghost `fm81_fractional_ghost_platonic.tex` template): here $c_{\mathcal{B}}^{\text{FL}}(k) = -(2k+3)(3k+1)/(k+3)$, derived by ordering the screening charges under W_3 -reflection before Sugawara subtraction. Under the Feigin–Frenkel dual $k \mapsto -k-6$, $c^{\text{FL}}(-k-6) = (2k+9)(3k+17)/(k+3)$ and the Koszul-conductor sum is

$$K_{\mathcal{B}}^{\text{FL}}(k) = c^{\text{FL}}(k) + c^{\text{FL}}(-k-6) = \frac{(2k+9)(3k+17) - (2k+3)(3k+1)}{k+3} = \frac{50(k+3)}{k+3} \equiv 50$$

in $\mathbb{Q}(k)$: the apparent pole at $k = -3$ is removable, and the Fateev–Lukyanov conductor is the polynomial constant 50. Both conventions therefore produce a level-independent polynomial-constant Koszul conductor; they differ only in numerical value (Arakawa $\equiv 196$, Fateev–Lukyanov $\equiv 50$) through the choice of central-charge normalisation, and they parametrise the same underlying W -algebra via a level reparametrisation. The specific numerical value 196 of Theorem 3.3 is Arakawa-convention-specific; the *constancy as a rational function of the level* is convention-independent. The Vol II cross-check in `bp_chain_level_strict_platonic.tex` uses FL consistently and confirms the polynomial constant 50.

Remark 3.8 (Convention choice rationale). Why FKR for the polynomial identity? Because FKR incorporates the Sugawara shift by $h^\vee(\mathfrak{sl}_3) = 3$ (the numerator $(k+1)$ shifts to $(k+1+2) = (k+3)$ under the $(k \mapsto -k-6)$ involution, producing the clean difference-of-squares $3 \cdot 8(k+3)/(k+3) = 24$ central to the proof). The FL convention suppresses this shift by absorbing it into the screening charge, producing instead the canonical W_3 -reflection $(2k+3)(3k+1)$ which has no difference-of-squares structure under the involution. The choice is not convention: the polynomial identity is a statement about a particular Chevalley–Shephard–Todd invariant of the $(\mathfrak{sl}_3, f_{\min})$ W -algebra pair, and FKR is the convention in which that invariant is manifestly visible.

3.3 Anomaly ratio and complementarity

Definition 3.9 (Anomaly ratio). The *anomaly ratio* of a chiral algebra $\mathcal{A}$ with modular characteristic κ and central charge c is $\rho(\mathcal{A}) := \kappa/c$.

Remark 3.10 (κ_T vs. ρ vs. ρ^{FKR} : naming disambiguation). Three numerical invariants in this paper share the informal epithet “anomaly ratio”; they are distinct functions of the algebra and must not be conflated.

- (a) **Modular characteristic** $\kappa_T(\mathcal{A})$ (c -dependent): the chain-level curvature coefficient appearing in Proposition 3.12. Family values: $\kappa_T(\text{Vir}_c) = c/2$; $\kappa_T(H_k) = k$ (Heisenberg at $c = 1$); $\kappa_T(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$ (affine Kac–Moody, cf. /); $\kappa_T(\mathcal{B}^k) = c(k)/2$.
- (b) **Reduced anomaly ratio** $\varrho(\mathcal{A}) := \kappa_T(\mathcal{A})/c$ (c -independent on any fixed family): $\varrho(\text{Vir}) = 1/2$; $\varrho(H_k) = k$ (since $c = 1$); $\varrho(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee c)$; $\varrho(\mathcal{B}^k) = 1/2$ (from $\kappa_T = c/2$). This is the invariant defined by Definition 3.9.
- (c) **FKR signed-sum ratio** $\varrho^{\text{FKR}}(\mathcal{A}) := \sum_i \varepsilon_i/(2h_i)$, summed over generator orbits with Cartan-grading signs (cf. Remark 3.11): $\varrho^{\text{FKR}}(\text{Vir}) = 1/(2 \cdot 2) = 1/4$ per generator, but with a single T -orbit and conventional doubling to match ϱ one obtains $1/2$; $\varrho^{\text{FKR}}(\mathcal{B}) = 1 - 4/3 + 1/2 = 1/6$ (eq. 3.1).

Relations: $\kappa_T(\mathcal{A}) = \varrho(\mathcal{A}) \cdot c$ holds identically by definition. The FKR signed-sum coincides with ϱ for pure Virasoro (both $1/2$) and diverges for composite algebras: for $\mathcal{B}$, $\varrho = 1/2$ but $\varrho^{\text{FKR}} = 1/6$ — the signed sum tracks BRST cohomology bookkeeping, not the κ_T/c ratio.

Convention used in this paper. The c -dependent modular characteristic κ_T is the operative invariant throughout (Proposition 3.12, the conductor Theorem 3.3, all downstream complementarity statements). The FKR signed-sum ratio ϱ^{FKR} appears only in equation (3.1) as a provenance check on the Kac–Roan–Wakimoto reduction, and is not used elsewhere.

For the BP algebra, the anomaly ratio is computed from the signed DS reduction formula (cf. the monograph, Computation comp:bp-kappa-three-paths): following the Feigin–Kac–Roan–Wakimoto convention (cf. the Bershadsky–Polyakov chapter of Volume I, and see Remark 3.11 below for the sign discipline): the J -generator (weight 1) contributes +1, the $G^\pm$ -ghost pair (weight $3/2$, one Cartan-graded fermion pair) contributes $-2/3 = -4/3$ combined, and the T -generator (weight 2) contributes $+1/2$. The result is

$$\varrho_{\mathcal{B}} = \frac{1}{1} - \frac{2}{3/2} + \frac{1}{2} = 1 - \frac{4}{3} + \frac{1}{2} = \frac{6 - 8 + 3}{6} = \frac{1}{6}. \quad (3.1)$$

Remark 3.11 (Signed-sum convention: why $G^\pm$ counts once). The FKR anomaly-ratio formula sums $\varepsilon_i/(2h_i)$ over *generator orbits* under the algebra’s natural grading, not over generators individually. The $G^\pm$ -pair, although realized as two fields of the same conformal weight, forms a *single* Cartan-graded fermion pair for the purposes of this sum: they carry opposite $U(1)$ -charges under J , and the contribution $-2/(2 \cdot 3/2) = -2/3$ per fermion doubles under the FKR orbit-count, yielding $-4/3$ combined. Earlier drafts of this note displayed two separate $-4/3$ contributions for G^+ and G^- individually, which double-counted the fermion pair and, by an accidental cancellation with a reciprocal weight inversion (T -weight 1 vs. 2, J -weight 2 vs. 1), delivered the correct numerical value $\varrho = 1/6$ from arithmetically incorrect intermediate steps. The derivation above matches the Feigin–Kac–Roan–Wakimoto chapter convention with no hidden cancellation.

Proposition 3.12 (Complementarity sum). *For the Bershadsky–Polyakov algebra with the FKR central charge:*

$$\kappa_T(k) + \kappa_T(k') = \frac{c(k)}{2} + \frac{c(k')}{2} = \frac{K_{\mathcal{B}}}{2} = 98.$$

*This is **nonzero**: the complementarity sum does not vanish. This is an instance of the anti-pattern from the monograph. For the Virasoro algebra, $\kappa + \kappa' = 13$; for the BP algebra, $\kappa_T + \kappa'_T = K_{\mathcal{B}}/2$.*

Remark 3.13 (in the non-principal landscape). The complementarity sum $\kappa + \kappa' = 0$ holds for affine Kac–Moody algebras (the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$ ensures antisymmetry around

o) and for free fields. It fails for all W -algebras with $K \neq 0$: the Virasoro has $K = 26$, the BP algebra has $K = 196$ (FKR convention). The general formula is $\kappa_T + \kappa'_T = K/2$. Writing $\kappa + \kappa' = 0$ for a W -algebra is always wrong.

4 Shadow obstruction tower

4.1 The T-line: class M

The T -line is the one-dimensional primary slice spanned by the Virasoro generator T . On this line, the shadow obstruction tower of $\mathcal{B}^k$ coincides with that of $\text{Vir}_{c(k)}$, because the shadow projections S_r on the T -line are determined entirely by the Virasoro self-OPE.

Theorem 4.1 (T-line shadow depth). *The shadow obstruction tower of $\mathcal{B}^k$ on the T -line has depth $r_{\max} = \infty$. The tower is class M (mixed). The shadow data on the T -line is:*

$$S_2 = \frac{c}{2}, \quad S_3 = 2, \quad S_4 = \frac{10}{c(5c + 22)}, \quad \Delta_T = \frac{40}{5c + 22}.$$

The critical discriminant $\Delta_T = 8\kappa_T S_4 = 40/(5c + 22)$ is nonzero for all $c \neq 0$, confirming that the shadow metric on the T -line is irreducible.

Proof. On the T -line, the MC equation projects to the single-line recursion with $\kappa = c/2$, $\alpha = S_3 = 2$, $S_4 = Q_{\text{Vir}}^{\text{contact}} = 10/(c(5c + 22))$ (the Virasoro quartic contact class). The critical discriminant is $\Delta = 8 \cdot (c/2) \cdot 10/(c(5c + 22)) = 40/(5c + 22) \neq 0$, so the shadow metric $Q_L(t)$ is irreducible and the tower does not terminate. $\square$

4.2 The J-line: class G

The J -line is the primary slice spanned by the $U(1)$ current J . The J -self-OPE is $J_{(1)}J = k_{\text{res}} = k + \frac{1}{2}$ (the residual $U(1)$ level), and all higher self-OPE modes vanish.

Theorem 4.2 (J-line shadow depth). *The shadow obstruction tower of $\mathcal{B}^k$ on the J -line has depth $r_{\max} = 2$. The tower is class G (Gaussian). The modular characteristic on the J -line is*

$$\kappa_J = k_{\text{res}} = k + \frac{1}{2}$$

where k_{res} is the residual $U(1)$ level. The cubic and quartic shadows vanish: $S_3^J = S_4^J = 0$.

Proof. The J -self-OPE is purely quadratic: $J_{(1)}J = k_{\text{res}}$ and $J_{(0)}J = 0$ (after $d \log$ absorption, the collision residue is k_{res}/z). The cubic shadow requires a composite-field contribution from the $J \cdot J$ OPE at mode (0) , which vanishes since $J_{(0)}J = 0$. Higher degrees vanish for the same reason: the shadow obstruction tower terminates at degree 2. $\square$

Remark 4.3 (Mixed depth within a single algebra). The Bershadsky–Polyakov algebra has shadow depth ∞ on the T -line and shadow depth 2 on the J -line. This is the first non-principal example of mixed shadow depth within a single algebra. In the principal landscape, each algebra has a single shadow class (Heisenberg is G on every line, $\widehat{\mathfrak{sl}}_N$ is L , Virasoro is M). The BP algebra demonstrates that non-principal W -algebras can exhibit different shadow classes on different generator lines. The global shadow class of $\mathcal{B}^k$ is M (the maximum over all lines), but the per-line classification is $(T : M, J : G, G^\pm : \text{fermionic})$.

4.3 Sigma-invariant

The sigma-invariant of a Koszul pair measures the “complementarity defect” at each degree:

$$\sigma^{(r)}(k) := S_r^T(c(k)) + S_r^T(c(k')), \quad k' = -k - 6.$$

For the T -line: $\sigma^{(2)} = c(k)/2 + c(k')/2 = K/2 = 98$. Higher degrees: $\sigma^{(3)} = 2 + 2 = 4$, $\sigma^{(4)} = 10/(c(5c + 22)) + 10/(c'(5c' + 22))$ where $c' = K - c$. These are generically nonvanishing.

Proposition 4.4 (Sigma non-vanishing). *For the Bershadsky–Polyakov algebra, $\sigma^{(r)} \neq 0$ for all $r \geq 2$ and generic k . In particular, the shadow complementarity sum does not vanish at any finite degree.*

Proof. At degree 2: $\sigma^{(2)} = K/2 = 98 \neq 0$. At degree 3: $\sigma^{(3)} = 4 \neq 0$. At degree 4: the quartic sigma-invariant is a nontrivial rational function of k . By the algebraicity theorem, all higher degrees are determined by degrees 2, 3, 4 via the single-line recursion, and the non-vanishing propagates. $\square$

5 The hook-type transport

5.1 The hook-type corridor

The hook-type corridor in type A consists of partitions $\lambda = (N - j, 1^j)$ of N . The transpose is $(N - j, 1^j)^t = (j + 1, 1^{N-j-1})$, another hook. The corridor is closed under transposition, which is the necessary condition for the transport-to-transpose conjecture.

Definition 5.1 (Self-transpose hooks). A hook partition $(N - j, 1^j)$ of N is *self-transpose* if and only if $(N - j, 1^j) = (j + 1, 1^{N-j-1})$, equivalently $N - j = j + 1$, i.e., $j = (N - 1)/2$. This requires N odd.

For $N = 3$: $j = 1, \lambda = (2, 1)$ (the BP algebra). For $N = 5$: $j = 2, \lambda = (3, 1, 1)$ (a hook-type W -algebra of $\mathfrak{sl}_5$). For $N = 2m + 1$: $j = m, \lambda = (m + 1, 1^m)$.

Theorem 5.2 (Hook self-duality). *For every odd $N = 2m + 1$ and $\lambda = (m + 1, 1^m)$, the W -algebra $W_k(\mathfrak{sl}_N, f_\lambda)$ is Koszul self-dual:*

$$(W_k(\mathfrak{sl}_N, f_\lambda))^! \cong W_{k'}(\mathfrak{sl}_N, f_\lambda)$$

with $k' = -k - 2N$, matching the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$ for $\mathfrak{sl}_N$ where $h^\vee(\mathfrak{sl}_N) = N$. The Koszul conductor is level-independent.

Proof. Self-transposition $\lambda = \lambda^t$ is immediate from Definition 5.1. The transport-to-transpose theorem of [2] then gives the Koszul duality, with the dual level determined by the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$; for $\mathfrak{sl}_N$ the dual Coxeter number is $h^\vee = N$, so $k' = -k - 2N$, consistent with Warning 5.3. For the Bershadsky–Polyakov case $N = 3$: $k' = -k - 6$, the familiar $\mathfrak{sl}_3$ duality. $\square$

Warning 5.3 (Dual Coxeter number). $h^\vee(\mathfrak{sl}_N) = N$. The Feigin–Frenkel involution is $k \mapsto -k - 2h^\vee = -k - 2N$. For $\mathfrak{sl}_3$: $k' = -k - 6$. The “ $-k - N$ ” in the theorem statement is WRONG if N denotes the rank; the correct shift is $-k - 2N$ where N is the rank of $\mathfrak{sl}_N$ (i.e., $h^\vee = N$ for type A_{N-1}).

5.2 The BP algebra as seed

The Bershadsky–Polyakov algebra occupies a distinguished position in the hook-type corridor: it is the simplest non-principal W -algebra that is both chirally Koszul (the bar spectral sequence collapses at E_2 by PBW universality of the universal algebra $W^k(\mathfrak{sl}_3, f_{(2,1)})$) and self-dual (the partition is self-transpose).

Computation 5.4 (Koszul invariants of $\mathcal{B}^k$). We tabulate the full set of modular Koszul invariants:

Invariant	Formula	Value
Central charge $c(k)$	$2 - 24(k+1)^2/(k+3)$	rational in k
Dual level k'	$-k - 6$	involution
Dual charge $c(k')$	$196 - c(k)$	by $K = 196$
Koszul conductor K	$c + c'$	196
$\kappa_T(k)$	$c(k)/2$	Virasoro sector
$\kappa_J(k)$	$(2k+1)/4$	$U(1)$ sector
Shadow depth (T -line)	∞	class $\mathcal{M}$
Shadow depth (J -line)	2	class G
$p_{\max}$	4	from $T_{(3)}T$
$k_{\max}$	3	$= p_{\max} - 1$

All entries are verified by the `bp_shadow_tower` and `bershadsky_polyakov_bar` compute modules (63 tests).

5.3 Comparison with Virasoro and W_3

The BP algebra sits between the Virasoro algebra and the principal W_3 algebra in the DS hierarchy of $\widehat{\mathfrak{sl}}_3$:

$$\widehat{\mathfrak{sl}}_3 \xrightarrow{\text{DS}_{(2,1)}} \mathcal{B}^k \xrightarrow{\text{further reduction}} \text{Vir}_{c(k)}.$$

The principal DS reduction of $\widehat{\mathfrak{sl}}_3$ produces W_3 (which has two generators T, W of weights 2, 3). The minimal DS reduction produces $\mathcal{B}^k$ (four generators of weights 1, 3/2, 3/2, 2). The Koszul conductors are:

$$K_{\text{Vir}} = 26, \quad K_{\mathcal{B}} = 196, \quad K_{W_3} = \frac{2 \cdot 8 \cdot (h^\vee)^2 \cdot \dim(\mathfrak{g})}{1} \text{ (computed from the } W_3 \text{ formula).}$$

The shadow class on the T -line is $\mathcal{M}$ for all three algebras. The distinguishing feature of $\mathcal{B}^k$ is the J -line, which is class G : the $U(1)$ current has only quadratic shadow data, and the tower terminates immediately. Neither the Virasoro (no J -line) nor W_3 (no J -line) exhibits this phenomenon.

5.4 Self-dual point

Proposition 5.5 (BP self-dual point). *The Bershadsky–Polyakov algebra is self-dual at $k = -3$, where $k' = -k - 6 = -3 = k$. At this level, $c(-3) = 2 - 24 \cdot 4/0$, which is singular: the self-dual point coincides with the*

critical level $k = -h^\vee = -3$. The self-dual central charge is therefore not a regular function value; the value $\kappa(\mathcal{B}^{-3}) = 49/3$ quoted elsewhere is the principal-value symmetric limit $\lim_{\varepsilon \rightarrow 0} \frac{1}{2} [\kappa(\mathcal{B}^{-3+\varepsilon}) + \kappa(\mathcal{B}^{-3-\varepsilon})]$ across the pole, not a function value (cf. Theorem 3.3). The “finite” self-dual point is at $c = 98$, but this value is not attained at any real level: the polynomial identity $K_{\mathcal{B}}(k) \equiv 196$ holds globally in $\mathbb{Q}(k)$ without any single level producing $c(\mathcal{B}^k) = 98$ as a regular value.

Proof. $c(k) = c(k')$ requires $c = K - c$, hence $c = K/2 = 98$. Solving $2 - 24(k+1)^2/(k+3) = 98$ reduces to $k^2 + 6k + 13 = 0$, whose discriminant is $36 - 52 = -16 < 0$. The roots $k = -3 \pm 2i$ are complex. Since $c(k) \leq 2$ for $k > -3$ and $c(k) \geq 194$ for $k < -3$ (with critical points at $k = -1$ and $k = -5$), the value $c = 98$ lies in the gap $(2, 194)$ and is never attained at a real level. The formal self-dual point $k = k' = -3$ is the critical level, where the DS reduction degenerates. $\square$

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